

CLOUD COMPUTING COURSE

Lecture23- Distributed Storage

ABSTRACT
Lecture23 Introduces distributed storage.

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Distributed Storage and Replication

Introduction

With the popularity of smartphones and sensors, the volume of data being stored and analyzed is growing rapidly, bringing about new business value. Distributed storage is often used to store massive volumes of data, in which multiple hard drives, solid-state drives, and other storage mechanisms are combined to be treated as a single storage. Storage capacity and performance can be improved by adding more servers. In addition, simultaneously storing r**eplicas** of a single data item on multiple servers increases the reliability of data.

Distributed Storage Definition:

Storing data on a multitude of standard servers (using servers local storage rather than using dedicated storage device), which behave as one storage system although data is distributed between these servers.

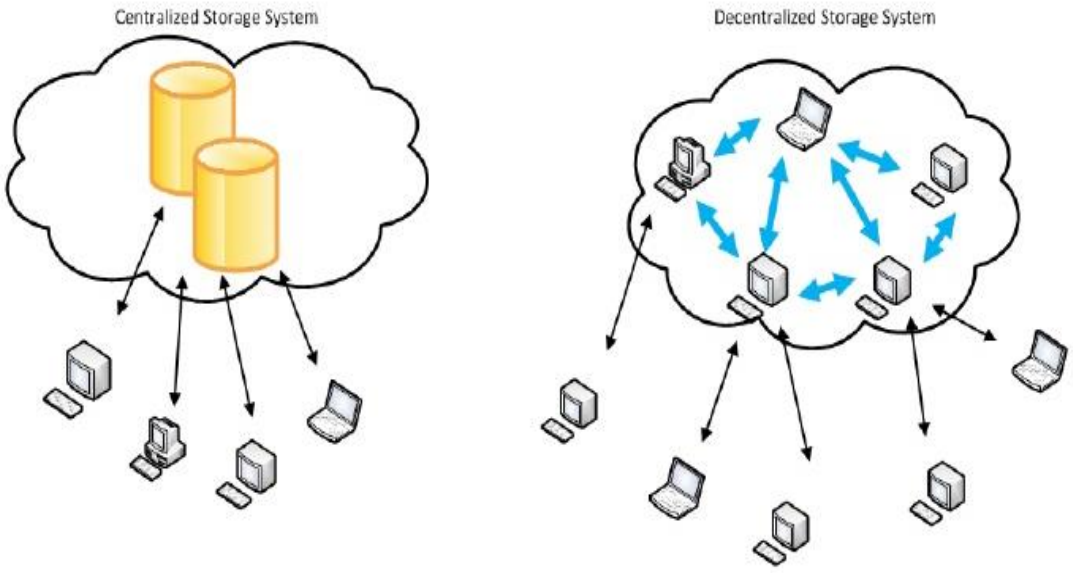
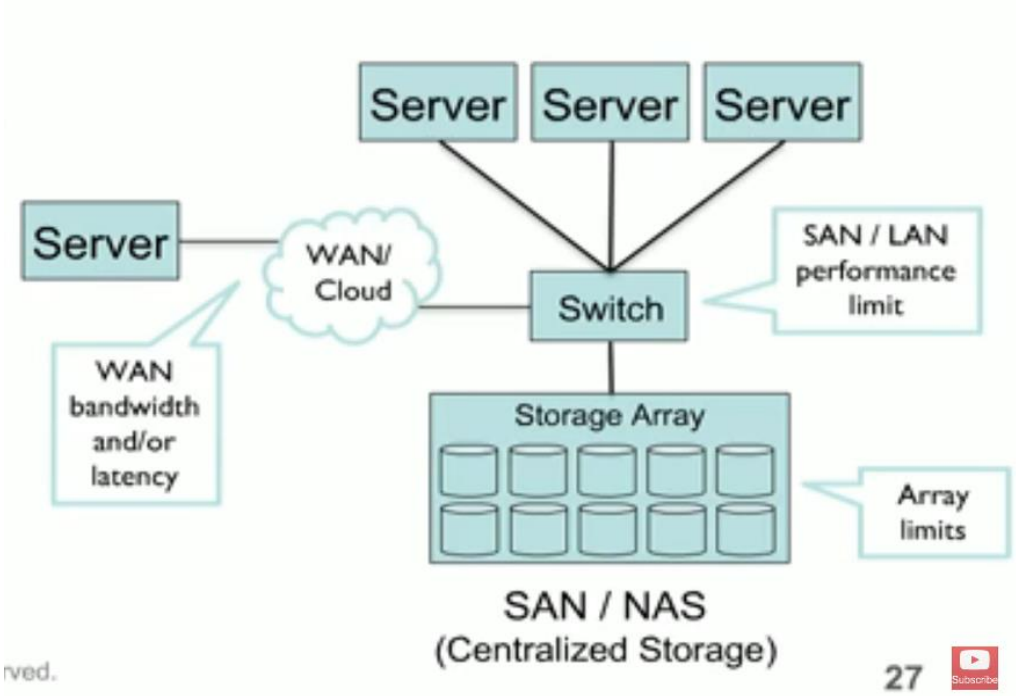
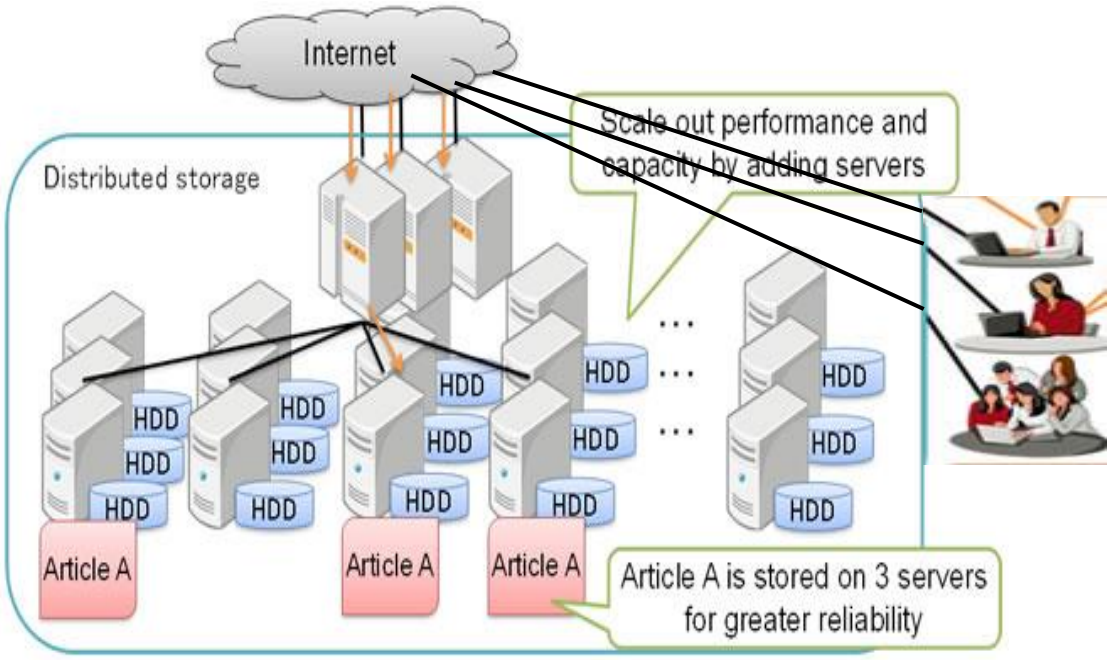


Fig1: Centralized Storage Vs Distributed Storage

Centralized Storage Vs Distributed Storage



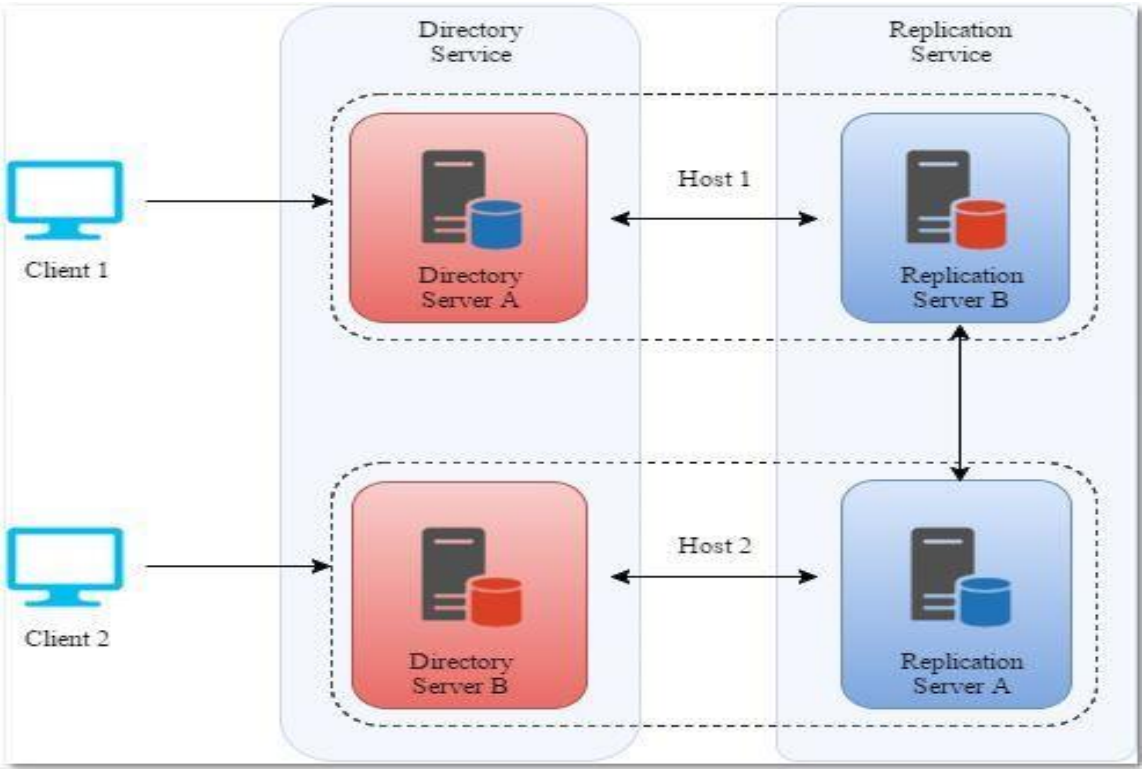
Centralized Storage

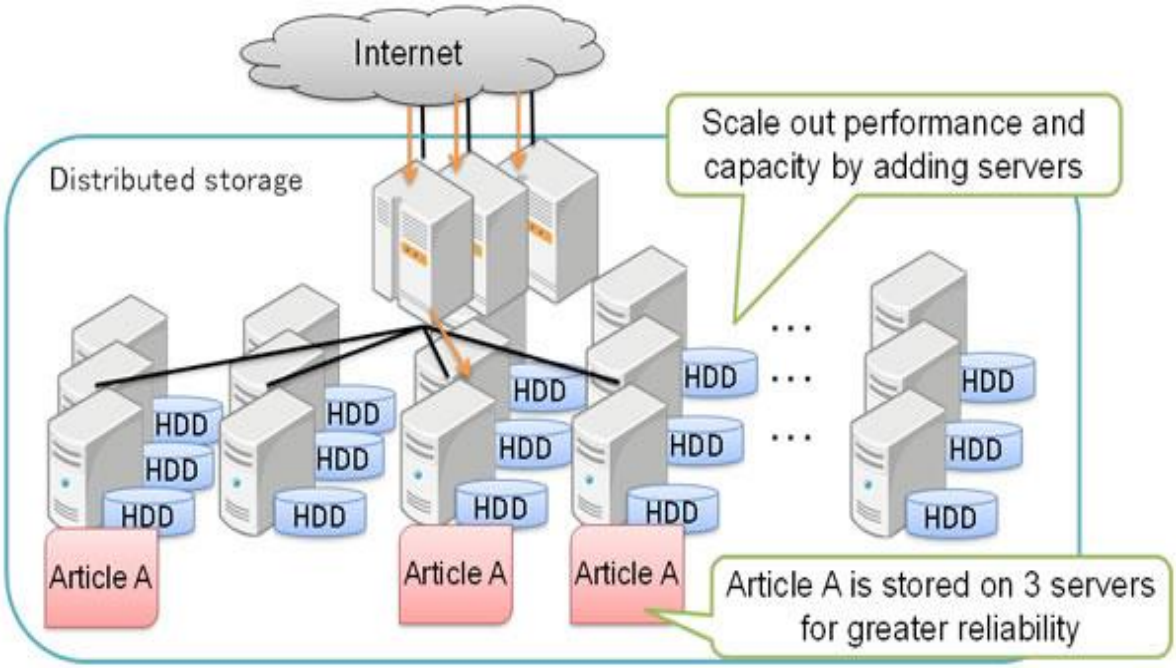


Distributed Storage

Replication Mechanism

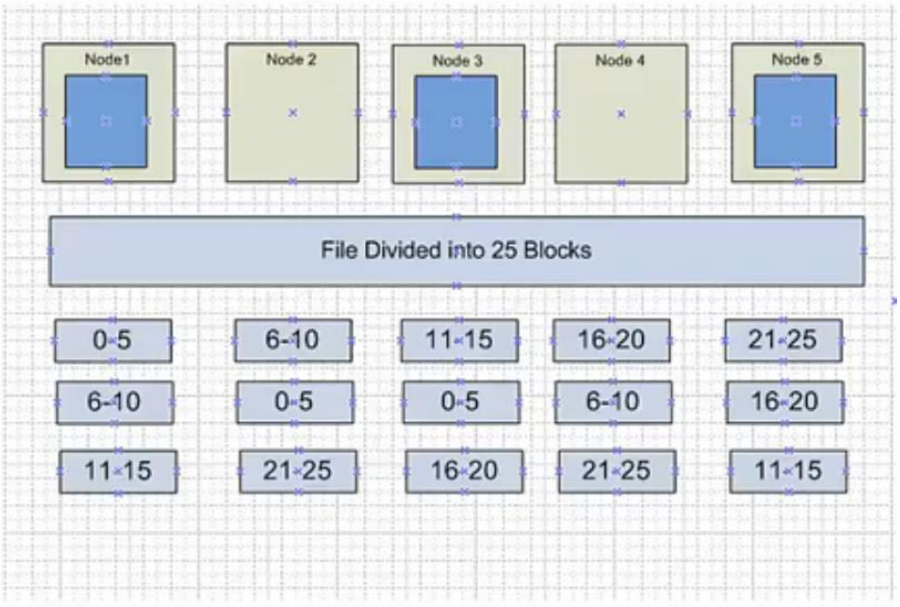
Replication in Cloud Computing refers to multiple storage of the same data to several different locations by usually synchronization of these data sources. Replication in Cloud Computing is partly done for backup and on the other hand to reduce response times, especially for reading data requests.



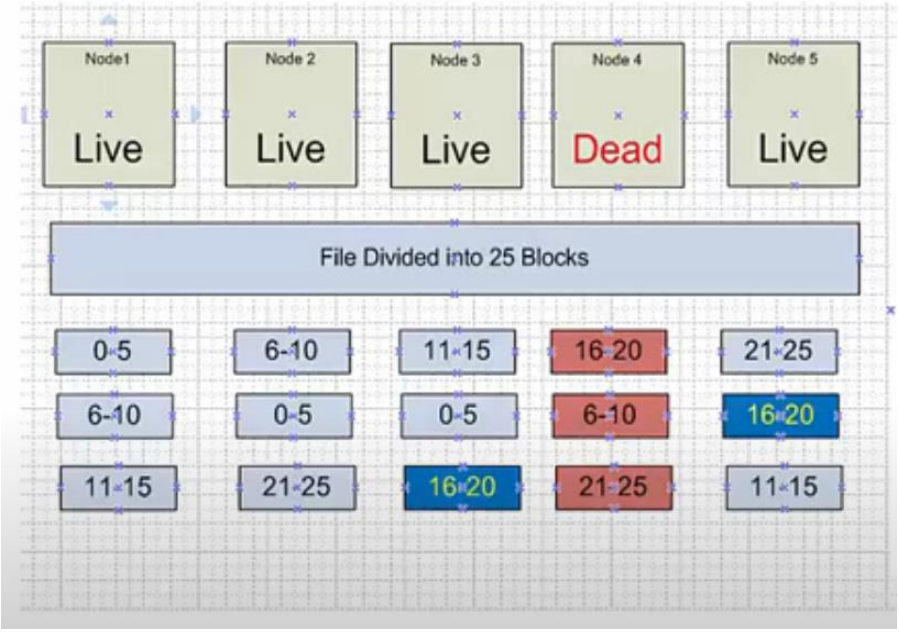


Distribution and Replication Deep Example

Saving file over multiple nodes (servers).



If one node fail→



You still have the data☺

Advantages of distributed Storage

FLEXIBILITY

This allows scaling by adding more servers and thus increasing capacity and performance linearly.

SPEED

In a distributed storage system, any server has CPU, RAM, drives and network interface and they all behave as one group. Therefore, any time you add a server you increase the total pool of resources and thus the speed of the entire system.

COST

Distributed storage uses standard servers, drives, and network, which are less expensive. It converges storage and compute, thus increasing the utilization of these standard servers. Consequently, less power, cooling, space, etc. are required in the data center. It is simpler to manage a distributed storage system, which means less staff would be required to run the IT infrastructure.

Distributed Storage Example

BigTable is a distributed storage system, which is designed by Google, just as its name, it is proposed to deal with data in large scale.